CURRICULUM VITAE



Dr.-Ing. Ramy Zeineldin

Nationality: Egyptian **Gender:** Male

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Scholar: scholar.google.com/citations?user=am3IeUYAAAAJ (*1,073 Citations | h-index: 14*)

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RESEARCH INTERESTS Artificial Intelligence, Computer Vision, Medical Image Analysis, and Surgical Robotics

ACADEMIC BACKGROUND

Jan 2024–Present **Assistant Professor,** *Computer Science and Engineering Department (CSE), Faculty of Electronic Engineering (FEE), Menoufia University (Egypt)*

Jun 2023–May 2025 **Postdoctoral Researcher,** *SPARC Lab, Artificial Intelligence in Biomedical Applications (AIBE), Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU), Erlangen (Germany)*
- AI and AR for Medical Applications; Medical and Surgical Robotics

Mar 2023–May 2023 **Postdoctoral Researcher,** *Computer Assisted Medicine research group (CaMed), Reutlingen University, Reutlingen (Germany)*
- Computational Imaging and AI for Medical Applications

Apr 2019–Feb 2023 **Doctoral Studies in Medical Computer Science,** *Health Robotics and Automation (HERA), Karlsruhe Institute of Technology (KIT), (Germany)*
- **Thesis Advisors:** Prof. Dr. Franziska Mathis-Ullrich, Prof. Dr.-Ing. Oliver Burgert

Feb 2018–Nov 2018 **Assistant Lecturer,** *CSE, FEE, Menoufia University (Egypt)*

Oct 2013–Jan 2018 **M.Sc. in Engineering Science - Computer Vision and Artificial Intelligence,** *FEE, Menoufia University (Egypt)*

Nov 2013–Jan 2018 **Teaching Assistant,** *CSE, FEE, Menoufia University (Egypt)*

Sep 2008–Jul 2013 **B.Sc. in Computer Science and Engineering,** *CSE, FEE, Menoufia University (Egypt)*

TEACHING EXPERIENCE AND SUPERVISION

Mar 2026–Present **Academic Supervisor**, *Menoufia Undergraduate Research Club (MURC), FEE, Menoufia University (Egypt)*

Nov 2025–Present **Assistant Professor**, *Reinforcement Learning, FEE, Menoufia University (Egypt)*

Jun 2025–Present **Assistant Professor**, *Computer Vision, Digital Multimedia and Virtual Reality, FEE, Menoufia University (Egypt)*

Sep 2025–Jan 2026 **Lecturer**, *Electronic Design Automation, Embedded Systems Design, Microprocessor Systems, Faculty of Engineering, October University for Modern Sciences and Arts University (MSA) (Egypt)*

Apr 2024–Oct 2025 **Lecturer**, *Empirical Research Methods in Medical Engineering, SPARC, AIBE, FAU University (Germany)*

Apr 2023–May 2025 **Lecturer**, *Seminar Surgical Robotics Hardware, Seminar Surgical Robotics Software, Seminar Cognitive Surgical Robotics, SPARC, AIBE, FAU University (Germany)*

Apr 2023–Feb 2025 **Co-supervision of one doctoral student and three MSc student theses**, *SPARC, AIBE, FAU University (Germany)*

Oct 2023–Jul 2024 **Co-supervision of one MSc research project**, *CaMed, Reutlingen University (Germany)*

Jun 2023–Aug 2023 **Assistant Lecturer**, *AI in Medical Robotics, SPARC, FAU University (Germany)*

Apr 2020–Feb 2022 **Assistant Lecturer**, *Computer Assisted Medicine Practical Course, CaMed, Reutlingen University (Germany)*

Dec 2017–Nov 2018	**Assistant Lecturer**, *Circuit Design using VHDL (FPGA), Computer Languages (C++), Microcontrollers and their applications), FEE, Menoufia University (Egypt)*
Nov 2013–Nov 2017	**Teaching Assistant**, *(Digital Logic Design, Computer Languages (C++), Computer Architecture, Microcontrollers and their applications, Digital System Design using VHDL, Advanced Computer Networks, Engineering Ethics), FEE, Menoufia University (Egypt)*
Mar 2020–May 2023	**Co-supervision of six MSc research projects**, *CaMed, Reutlingen University (Germany)*
Sep 2014–Jul 2018	**Co-supervision of Bachelor-Theses Students**, *Faculty of Electronic Engineering, Menoufia University (Egypt)*

WORK AND PROJECT EXPERIENCE

Jan 2018–Mar 2018	**Internship**, *Embedded systems, Information Technology Institute (ITI), Cairo (Egypt)*
Jun 2017–Sep 2017	**Internship**, *Cloud and Mobile Computing using Watson, IBM, Cairo (Egypt)*
Sep 2012–Jan 2015	**Co-founder, R&D engineer**, *Mubser, LLC, Cairo (Egypt)*

SELECTED PUBLICATIONS

[1] **Zeineldin, R. A.**, Karar, M. E., Coburger, J., Wirtz, C. R., & Burgert, O. (2020). DeepSeg: deep neural network framework for automatic brain tumor segmentation using magnetic resonance FLAIR images. *International journal of computer assisted radiology and surgery*, 15(6), 909-920. https://doi.org/10.1007/s11548-020-02186-z

[2] **Zeineldin, R. A.,** Karar, M. E., Elshaer, Z., Coburger, J., Wirtz, C. R., Burgert, O., & Mathis-Ullrich, F. (2024). Explainable hybrid vision transformers and convolutional network for multimodal glioma segmentation in brain MRI. *Nature Scientific Reports*, *14*(1), 3713. https://doi.org/10.1038/s41598-024-54186-7

[3] **Zeineldin, R. A.**, Karar, M. E., Elshaer, Z., Wirtz, C. R., Burgert, O., & Mathis-Ullrich, F. (2022). Explainability of deep neural networks for MRI analysis of brain tumors. *International Journal of Computer Assisted Radiology and Surgery*, 1-11. https://doi.org/10.1007/s11548-022-02619-x

[4] Matasyoh, N. M., Delev, D., Masalha,W., Mathis-Ullrich, F., & **Zeineldin, R. A.** (2025). NeuroLens: Enhancing Surgical Education with Real-time Organ Localization using Natural Language Commands. *International journal of computer assisted radiology and surgery,* 20(8), 1623-1632. https://doi.org/10.1007/s11548-025-03463-5

MEMBERSHIPS

2020–Today	Medical Image Computing and Computer Assisted Intervention (**MICCAI**)
2020–Today	Computer Assisted Radiology and Surgery (**CARS**)
2019–Today	Institute of Electrical and Electronics Engineers (**IEEE**)
2013–Today	Engineering Syndicate of Egypt (**EE**)
2023–2024	Association for Computing Machinery (**ACM**)
2019–2023	German Society for Computer- and Robot-Assisted Surgery (**CURAC**)
2020–2023	German Informatics Society (**GI**) (German: Gesellschaft für Informatik)
2020–2023	IEEE Engineering in Medicine & Biology Society (**EMBS**)

RECOGNITIONS (SELECTED)

Jul 2024	**Best Doctoral Dissertation**, *Erika und Dr. Wolfgang Eichelberger-Stiftung, KIT, Karlsruhe*
Sep 2023	**Top 200 young researchers worldwide,** *Heidelberg Laureate Forum (HLF), Heidelberg*
Sep 2022	**Winner**, *Brain Tumor Segmentation (BraTS) 2022 Challenge among more than 1000 international teams and corporates (e.g. Nvidia and Google), MICCAI (Singapore)*
Sep 2022	**Top 5**, *Brain Tumor Sequence Registration (BraTS-Reg) Challenge, MICCAI (Singapore)*
Sep 2021	**Top 10**, *BraTS 2021 Challenge, MICCAI, Strasbourg (France)*
Sep 2020	**Best poster award** *at the 19th Annual Conference of the German Society for Computer- and Robot-Assisted Surgery (CURAC), Hamburg (Germany)*
Sep 2020	**Top 200 young researchers worldwide**, *HLF, Heidelberg (online)*
Apr 2018-Mar 2023	**Holder** *of the German Academic Exchange Service (DAAD) scholarship (Germany)*

May 2014	**Finalist**, *MIT Enterprise Forum, Pan Arab Region (Lebanon)*
Nov 2013	**Winner**, *Idea to Product Global Competition, Sao Paulo (Brazil)*
Sep 2013	**Finalist**, *Made in Egypt (MIE) Competition, IEEE Gold (Egypt)*
June 2013	**Winner**, *Intel Egypt Award, Intel (Egypt)*
July 2013	**1st Rank**, *Computer Science and Engineering Dept., FEE, Menoufia University (Egypt)*

EXTRA-CURRICULAR ACTIVITIES

Nov 2023–Today	**Fellow and Mentor**, *CaMed, Reutlingen University (Germany)*
Feb 2023–Today	**Alumni**, *DAAD (Germany)*
Oct 2022–Today	**Alumni and Mentor**, *AlumNode, the Klaus Tschira Stiftung (KTS) (Germany)*
2019–Today	**Reviewer**, *several scientific journals (e.g., MedIA, Journal of Medical Systems, IEEE JBHI, IEEE Access, Nature Scientific Reports, BMC Medical Imaging, at Auto) and conferences (e.g., MICCAI, CURAC, CARS, IJCARS)*
Sep–Nov 2023	**Co-organizer,** *the AI in Biomedical Engineering Conference, FAU, Erlangen (Germany)*
Sep 2019	**Co-organizer,** *CURAC 2019, Reutlingen (Germany)*
Feb 2018 Nov 2018	**Co-founder,** *two laboratories (Computer Vision and Biometrics, and Printed Circuit Board (PCB)), FEE, Menoufia University (Egypt)*
Mar 2016–Jun 2018	**Instructor & Judge,** *FEE Robotics Club, Menoufia University (Egypt)*
Mar 2015–Apr 2016	**Instructor & Mentor,** *IEEE From School to College (STC), Menoufia University (Egypt)*
Sep 2014–Jul 2016	**Mentor & Supervisor,** *Embedded Systems Foundation, Menoufia University (Egypt)*
Nov 2010– Aug 2011	**Member of the Students' Union,** *FEE, Menoufia University (Egypt)*